

MSTAR PRACTICE WORKSHEET**Lesson 7-3: Understand Absolute Value of Rational Numbers**

Grade 6 Mathematics · Standard 6.NOS.8

These are MSTAR-style questions written for this curriculum to rehearse the format. They are not official Maryland assessment items. MSTAR — Maryland's new state test, first given in spring 2027 — uses the same kinds of questions you see here: selected response, select-all, two-part evidence questions, and written responses.

Part A — Skills Check

One point each.

Question 1 **SELECTED RESPONSE — CORRECT: D**

What is the absolute value of -9?

- ☐ A -9
- ☐ B 0
- ☐ C 1
- ☒ D 9

Absolute value is the distance from zero. -9 is 9 units from zero, so $|-9| = 9$.

Question 2 **SELECTED RESPONSE — CORRECT: D**

Which integer has the GREATEST absolute value: -3, 7, -5, 2?

- ☐ A -5
- ☐ B -3
- ☐ C 2
- ☒ D 7

$|-3| = 3$, $|7| = 7$, $|-5| = 5$, $|2| = 2$. The greatest absolute value is 7.

Question 3 SELECTED RESPONSE — CORRECT: C

What is $|0|$?

- ☐ A 1
- ☐ B -1
- ☒ C 0
- ☐ D There is no answer

Zero is 0 units from itself on the number line, so $|0| = 0$.

Question 4 SELECTED RESPONSE — CORRECT: B

What is $|-2.5|$?

- ☐ A 0
- ☒ B 2.5
- ☐ C -2.5
- ☐ D 25

Absolute value is distance from zero, and -2.5 sits 2.5 units from zero, so $|-2.5| = 2.5$.

Question 5 SELECTED RESPONSE — CORRECT: B

What is $|-3/4|$?

- ☐ A 0
- ☒ B $3/4$
- ☐ C $-3/4$
- ☐ D $4/3$

$|-3/4| = 3/4$. Absolute value is distance from zero, and $-3/4$ sits $3/4$ of a unit from 0 — fractions work exactly like integers here.

Question 6 **SELECTED RESPONSE — CORRECT: A**

Which statement about absolute value is TRUE?

- ☒ **A** $|-4| = |4|$ because both are 4 units from zero
- ☐ **B** $|-4| = -4$ because the number is negative
- ☐ **C** $|-4| = 0$ because negatives cancel out
- ☐ **D** $|4| > |-4|$ because positive is always bigger

Absolute value measures distance from zero. Both -4 and 4 are 4 units from zero, so $|-4| = |4| = 4$.

Part B — Test-Format Questions

Scoring notes and distractor rationales below each item.

Question 7 **TWO-PART QUESTION****Part A — correct answer: C**

What is the value of $|-12|$?

- ☐ **A** 0
- ☐ **B** $1/12$
- ☒ **C** 12
- ☐ **D** -12

Absolute value is distance from zero, and distance is never negative. -12 sits 12 units from zero, so $|-12| = 12$.

- **A:** The bars ask how FAR from zero, not the difference with itself. Negative 12 sits 12 units away.
- **B:** The bars mean distance from zero, not a reciprocal.
- **D:** Absolute value is a distance from zero, and a distance is never negative.

Part B — correct answer: D

Why can an absolute value never be negative?

- ☐ **A** Negative signs cancel out automatically.
- ☐ **B** The bars change every number to a positive.
- ☐ **C** Absolute value always adds the number to itself.
- ☒ **D** It measures distance from zero, and a distance is never negative.

$|a|$ is defined as how FAR a is from zero, regardless of direction. Distance has no direction, so it is always zero or positive.

Question 8

WRITTEN RESPONSE · 2 POINTS

Type II Reasoning: Bigger Absolute Value, Bigger Number?

Owen compared -8 and -3 . He wrote: 'Since $|-8| = 8$ and $|-3| = 3$, and $8 > 3$, then $-8 > -3$.'

Explain Owen's misconception and state the correct comparison.

Model answer: $-8 < -3$. Absolute value tells you how far a number is from zero, not which is greater. A negative number with a larger absolute value sits further left, so it is SMALLER. Owing \$8 is worse than owing \$3.

2 points	Student explains that absolute value measures distance and ignores direction, that on the number line -8 lies further LEFT than -3 , and concludes $-8 < -3$.
1 point	Student states $-8 < -3$ without explaining why absolute value cannot be used to compare.
0 points	Student agrees with Owen or gives an unrelated response.

Part C — Show Your Thinking

Model answer and scoring guidance.

Question 9

WRITTEN RESPONSE · 2 POINTS

A submarine is at -150 meters and a bird is flying at $+150$ meters. A student says the submarine is closer to sea level because -150 is less than 150 . Is the student correct? Explain using absolute value.

Model answer: The student is not correct. Absolute value measures distance from sea level, and $|-150| = 150$ while $|150| = 150$, so the submarine and the bird are exactly the same distance from 0. The comparison $-150 < 150$ describes position — below versus above sea level — not closeness to it.

Award 2 points for a complete explanation with correct mathematics, 1 point for a correct answer with thin reasoning, 0 for an unrelated response.

Answer Key. Score written responses with the rubric and guidance above; award partial credit per the 1-point rows.